

$$\cos \frac{\pi}{3} = \cos 60^\circ = \frac{1}{2}$$

$$\begin{aligned}\text{所求} &= \frac{\sqrt{3}}{2} \times \frac{\sqrt{3}}{2} + 1 \times (-1) + \frac{1}{2} \times \frac{1}{2} \\ &= \frac{3}{4} + (-1) + \frac{1}{4} = 0\end{aligned}$$

$$2. \sin 120^\circ = \sin(180^\circ - 60^\circ) = \sin 60^\circ = \frac{\sqrt{3}}{2}$$

$$\cos 210^\circ = \cos(180^\circ + 30^\circ) = -\cos 30^\circ = -\frac{\sqrt{3}}{2}$$

$$\tan 315^\circ = \tan(360^\circ - 45^\circ) = -\tan 45^\circ = -1$$

$$\begin{aligned}\cos(-300^\circ) &= \cos 300^\circ = \cos(360^\circ - 60^\circ) \\ &= \cos 60^\circ = \frac{1}{2}\end{aligned}$$

$$\text{所求} = \frac{\sqrt{3}}{2} + \left(-\frac{\sqrt{3}}{2}\right) + (-1) + \frac{1}{2} = -\frac{1}{2}$$

$$\begin{aligned}3. \text{ 所求} &= \frac{\cos \theta}{\sin \theta} + \frac{\tan \theta}{-\tan \theta} - \frac{-\cos \theta}{-\sin \theta} \\ &= \frac{\cos \theta}{\sin \theta} + (-1) - \frac{\cos \theta}{\sin \theta} = -1\end{aligned}$$

4. 圖形是 $y = \cos x$ 以 x 軸為基準上下拉伸 3 倍
即 $y = 3\cos x$ ，故選甲

5. $\because y = \sin x$ 的週期是 2π
而 $y = \sin 2x$ 的 x 係數是 2

$$\therefore y = \sin 2x \text{ 的週期為 } \frac{2\pi}{2} = \pi, \text{ 即 } a = \pi$$

而 $y = 2\sin x$ 的週期與 $y = \sin x$ 的相等
都是 2π ，即 $b = 2\pi$

$$\text{故 } a + 2b = \pi + 2 \times (2\pi) = \pi + 4\pi = 5\pi$$

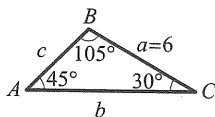
6. $\angle A = 180^\circ - \angle B - \angle C = 180^\circ - 105^\circ - 30^\circ = 45^\circ$
由正弦定理

$$\frac{a}{\sin A} = \frac{c}{\sin C}$$

$$\Rightarrow \frac{6}{\sin 45^\circ} = \frac{c}{\sin 30^\circ}$$

$$\Rightarrow \sin 45^\circ \times c = \sin 30^\circ \times 6 \Rightarrow \frac{\sqrt{2}}{2} \times c = \frac{1}{2} \times 6$$

$$\Rightarrow c = \frac{6}{\frac{\sqrt{2}}{2}} = \frac{6\sqrt{2}}{2} = 3\sqrt{2}$$



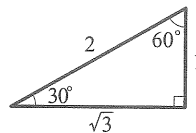
$$7. \because a : b : c = \sin A : \sin B : \sin C = 1 : \sqrt{3} : 2$$

$$\therefore \angle A = 30^\circ, \angle B = 60^\circ, \angle C = 90^\circ$$

$$\text{則 } \sin A = \sin 30^\circ = \frac{1}{2}, \cos B = \cos 60^\circ = \frac{1}{2},$$

$$\sin C = \sin 90^\circ = 1$$

$$\text{故 } \sin A + \cos B + \sin C = \frac{1}{2} + \frac{1}{2} + 1 = 2$$



$$8. \text{ 令 } a = \overline{BC}, b = \overline{AC}, c = \overline{AB}$$

由餘弦定理

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$= 4^2 + 2^2 - 2 \times 4 \times 2 \times \cos 120^\circ$$

$$= 16 + 4 - 2 \times 4 \times 2 \times \left(-\frac{1}{2}\right) = 28$$

$$\text{則 } a = \sqrt{28} = 2\sqrt{7}, \text{ 故 } \overline{BC} = 2\sqrt{7}$$

$$9. \overline{AD} = \sqrt{3^2 + 1^2} = \sqrt{10}, \overline{AB} = \sqrt{3^2 + 3^2} = \sqrt{18} = 3\sqrt{2}$$

由餘弦定理，在 $\triangle ABD$ 之中

$$\cos \theta = \frac{\overline{AB}^2 + \overline{AD}^2 - \overline{BD}^2}{2 \times \overline{AB} \times \overline{AD}}$$

$$= \frac{(3\sqrt{2})^2 + (\sqrt{10})^2 - 2^2}{2 \times 3\sqrt{2} \times \sqrt{10}} = \frac{24}{12\sqrt{5}}$$

$$= \frac{2}{\sqrt{5}} = \frac{2\sqrt{5}}{5}$$

$$10. \text{ 令 } s = \frac{1}{2} \times (70 + 80 + 90) = 120$$

由海龍公式

$$\begin{aligned}\text{面積} &= \sqrt{120 \times (120 - 70) \times (120 - 80) \times (120 - 90)} \\ &= \sqrt{120 \times 50 \times 40 \times 30} = 1200\sqrt{5}\end{aligned}$$

⑤ 單元 3 平面向量

(主題 1~8：向量及其基本運算)

答案

1. 4

2. 8

3. (5, -5)

4. -3

5. (-2, 3)

6. 3

7. 2

8. $\left(\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$

9. $\left(\frac{3}{5}, -\frac{4}{5}\right)$

10. -4



- $\overrightarrow{AB} = B - A = (8, -1) - (5, 3) = (3, -4)$
 $|\overrightarrow{AB}| = \sqrt{3^2 + (-4)^2} = \sqrt{25} = 5$
 則 $m = 3$, $n = -4$, $k = 5$
 故 $m + n + k = 3 + (-4) + 5 = 4$
- $\because \overrightarrow{a} = \overrightarrow{b}$
 $\therefore x + y = 7 \cdots \cdots \textcircled{1}$, $x - y = 5 \cdots \cdots \textcircled{2}$
 $\textcircled{1} + \textcircled{2} : 2x = 12 \Rightarrow x = 6$
 $x = 6$ 代入 $\textcircled{1} : 6 + y = 7 \Rightarrow y = 1$
 故 $x + 2y = 6 + 2 \times 1 = 8$
- $\because \overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{AC}$
 而 $\overrightarrow{AC} = C - A = (4, -2) - (-1, 3) = (5, -5)$
 $\therefore \overrightarrow{AB} + \overrightarrow{BC} = (5, -5)$
- $\because \overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CD} + \overrightarrow{DE} = \overrightarrow{AE}$
 $\therefore \overrightarrow{AE} = (m, n)$
 而 $\overrightarrow{AE} = E - A = \left(-\frac{23}{4}, -\frac{10}{3}\right) - \left(\frac{1}{3}, -\frac{1}{4}\right)$
 $= \left(-\frac{73}{12}, -\frac{37}{12}\right)$
 故 $m = -\frac{73}{12}$, $n = -\frac{37}{12}$
 因此 $m - n = -\frac{73}{12} - \left(-\frac{37}{12}\right) = -3$
- $\overrightarrow{AB} = B - A = (-1, 5) - (3, 2) = (-4, 3)$
 $\overrightarrow{BC} = C - B = (9, -4) - (-1, 5) = (10, -9)$
 $\overrightarrow{CA} = A - C = (3, 2) - (9, -4) = (-6, 6)$
 $\overrightarrow{AB} + 2\overrightarrow{BC} + 3\overrightarrow{CA}$
 $= (-4, 3) + 2(10, -9) + 3(-6, 6)$
 $= (-4, 3) + (20, -18) + (-18, 18) = (-2, 3)$
- $\because \overrightarrow{a}$ 、 $\overrightarrow{b}$ 方向相同且 $30 \div 10 = 3$
 $\therefore \overrightarrow{b} = 3\overrightarrow{a}$, 故 $n = 3$
- $3\overrightarrow{a} - 2\overrightarrow{b} = 3(1, 2) - 2(2, 3)$
 $= (3, 6) - (4, 6) = (-1, 0)$
 故 $r = -1$, $s = 0$, 因此 $s - 2r = 0 - 2 \times (-1) = 2$

- $|\overrightarrow{a}| = \sqrt{(\sqrt{3})^2 + (-1)^2} = 2$
 $\therefore |\overrightarrow{u}| = 1$
 $\therefore \overrightarrow{a}$ 的長度是 $\overrightarrow{u}$ 的 2 倍
 而 $\overrightarrow{u}$ 、 $\overrightarrow{a}$ 方向相同
 故 $\overrightarrow{u} = \frac{1}{2}\overrightarrow{a} = \frac{1}{2}(\sqrt{3}, -1) = \left(\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$
- $\overrightarrow{AB} = B - A = (-1, 3) - (2, -1) = (-3, 4)$
 $|\overrightarrow{AB}| = \sqrt{(-3)^2 + 4^2} = 5$
 $\therefore \overrightarrow{u}$ 與 $\overrightarrow{AB}$ 方向相反且 $|\overrightarrow{u}| = 1$
 $\therefore \overrightarrow{u} = -\frac{1}{5}\overrightarrow{AB} = -\frac{1}{5}(-3, 4) = \left(\frac{3}{5}, -\frac{4}{5}\right)$
- $\overrightarrow{a} + \overrightarrow{b} = (5t, 3) + (60, -9) = (5t + 60, -6)$
 $\overrightarrow{a} - \overrightarrow{b} = (5t, 3) - (60, -9) = (5t - 60, 12)$
 $\therefore (\overrightarrow{a} + \overrightarrow{b})$ 與 $(\overrightarrow{a} - \overrightarrow{b})$ 平行
 $\therefore \frac{5t + 60}{5t - 60} = \frac{-6}{12} \Rightarrow \frac{t + 12}{t - 12} = \frac{-1}{2}$
 $\Rightarrow 2 \times (t + 12) = -1 \times (t - 12)$
 $\Rightarrow 2t + 24 = -t + 12 \Rightarrow 3t = -12$
 $\Rightarrow t = -4$

⑥ 單元 3 平面向量

(主題 9~17：向量的內積、應用)

答案

- | | |
|----------------|-------------|
| 1. 5 | 2. 3 |
| 3. $\sqrt{31}$ | 4. 6 |
| 5. 3 | 6. -6 |
| 7. 60 | 8. 10 |
| 9. 52 | 10. (14, 2) |

- $\overrightarrow{a} \cdot \overrightarrow{b} = |\overrightarrow{a}| \times |\overrightarrow{b}| \times \cos 60^\circ = 5 \times 2 \times \frac{1}{2} = 5$
- $\overrightarrow{b} \cdot (5\overrightarrow{a} - 2\overrightarrow{b}) = \overrightarrow{b} \cdot (5\overrightarrow{a}) - \overrightarrow{b} \cdot (2\overrightarrow{b})$
 $= 5\overrightarrow{a} \cdot \overrightarrow{b} - 2|\overrightarrow{b}|^2$
 $= 5 \times 15 - 2 \times 6^2 = 75 - 72 = 3$

$$3. \quad \vec{a} \cdot \vec{b} = |\vec{a}| \times |\vec{b}| \times \cos 30^\circ = 4 \times \sqrt{3} \times \frac{\sqrt{3}}{2} = 6$$

$$\begin{aligned} |\vec{a} + \vec{b}|^2 &= |\vec{a}|^2 + 2\vec{a} \cdot \vec{b} + |\vec{b}|^2 \\ &= 4^2 + 2 \times 6 + (\sqrt{3})^2 = 31 \end{aligned}$$

$$\text{故 } |\vec{a} + \vec{b}| = \sqrt{31}$$

$$4. \quad \vec{AB} = B - A = (5, -2) - (-1, 1) = (6, -3)$$

$$\vec{AC} = C - A = (1, 3) - (-1, 1) = (2, 2)$$

$$\vec{AB} \cdot \vec{AC} = (6, -3) \cdot (2, 2) = 6 \times 2 + (-3) \times 2 = 6$$

$$\begin{aligned} 5. \quad \vec{a} \cdot \vec{b} &= (1, k+1) \cdot (2k, 3) \\ &= 1 \times 2k + (k+1) \times 3 \\ &= 2k + (3k+3) = 5k+3 \end{aligned}$$

$$\text{則 } 5k+3=18 \Rightarrow 5k=15 \Rightarrow k=3$$

$$\begin{aligned} 6. \quad \because \vec{a} \perp \vec{b} \\ \therefore \vec{a} \cdot \vec{b} &= 0 \end{aligned}$$

$$\begin{aligned} \text{而 } \vec{a} \cdot \vec{b} &= (3, 4) \cdot (x, -9) \\ &= 3 \times x + 4 \times (-9) = 3x - 36 \end{aligned}$$

$$\text{則 } 3x - 36 = 0 \Rightarrow 3x = 36 \Rightarrow x = 12$$

$$\therefore \vec{b} \parallel \vec{c}$$

$$\begin{aligned} \therefore \frac{x}{-8} = \frac{-9}{y} &\Rightarrow \frac{12}{-8} = \frac{-9}{y} \Rightarrow \frac{3}{-2} = \frac{-9}{y} \\ &\Rightarrow 3y = 18 \Rightarrow y = 6 \end{aligned}$$

$$\text{故 } y - x = 6 - 12 = -6$$

$$7. \quad \begin{vmatrix} 5 & 3 \\ 45 & 30 \end{vmatrix} = 5 \times 30 - 3 \times 45 = 150 - 135 = 15$$

$$\begin{vmatrix} 7 & -4 \\ 6 & 3 \end{vmatrix} = 7 \times 3 - (-4) \times 6 = 21 - (-24) = 45$$

$$\text{則 } m=15, n=45, \text{ 故 } m+n=15+45=60$$

$$8. \quad \vec{AB} = B - A = (98, 89) - (90, 80) = (8, 9)$$

$$\vec{AC} = C - A = (94, 87) - (90, 80) = (4, 7)$$

$$\begin{aligned} \triangle ABC \text{ 面積} &= \frac{1}{2} \begin{vmatrix} 8 & 9 \\ 4 & 7 \end{vmatrix} = \frac{1}{2} \times |8 \times 7 - 9 \times 4| \\ &= \frac{1}{2} \times |20| = \frac{1}{2} \times 20 = 10 \end{aligned}$$

$$9. \quad \text{由柯西不等式: } (2^2 + 3^2)(x^2 + y^2) \geq (2x + 3y)^2$$

$$\Rightarrow 13(x^2 + y^2) \geq 26^2 \Rightarrow x^2 + y^2 \geq 52$$

故 $x^2 + y^2$ 的最小值為 52

10. $\vec{a}$ 在 $\vec{b}$ 上的正射影:

$$\begin{aligned} \left(\frac{\vec{a} \cdot \vec{b}}{|\vec{b}|^2} \right) \vec{b} &= \frac{(12, 16) \cdot (7, 1)}{(\sqrt{7^2 + 1^2})^2} (7, 1) \\ &= \frac{12 \times 7 + 16 \times 1}{50} (7, 1) \\ &= 2(7, 1) = (14, 2) \end{aligned}$$

7 單元 4 式的運算

(主題 1~10: 多項式的四則運算)

答案

- | | |
|--------------------------|-------------------------------|
| 1. 2 | 2. -15 |
| 3. $9x^4 - x^3$ | 4. $64x^3 - 140x^2 + 88x - 2$ |
| 5. 13 | 6. -7 |
| 7. 31 | 8. 7 |
| 9. $2x^3 + x^2 - 3x + 2$ | 10. 10 |

$$1. \quad \because f(x) = g(x)$$

$\therefore$ 同次項的係數相等

$$\text{故 } -2 = b - 1, a + 1 = -1, 5 = c$$

$$\Rightarrow b = -1, a = -2, c = 5$$

$$\text{因此 } a + b + c = -2 + (-1) + 5 = 2$$

$$\begin{aligned} 2. \quad f(x) + g(x) &= (3x^2 - bx + c) + (ax^2 + 5x - 1) \\ &= (a+3)x^2 + (-b+5)x + (c-1) \end{aligned}$$

$\therefore f(x) + g(x)$ 為零多項式

$$\therefore a+3=0, -b+5=0, c-1=0$$

$$\text{故 } a=-3, b=5, c=1$$

$$\text{因此 } abc = -3 \times 5 \times 1 = -15$$

$$3. \quad f(x) = (3x^2)^2 - 8^2 = 9x^4 - 64$$

$$g(x) = (x-4)(x^2 + 4 \times x + 4^2) = x^3 - 4^3 = x^3 - 64$$

$$\text{故 } f(x) - g(x) = (9x^4 - 64) - (x^3 - 64) = 9x^4 - x^3$$

$$4. \quad f(x) = (2x)^2 - 2 \times 2x \times 5 + 5^2 = 4x^2 - 20x + 25$$

$$\begin{aligned} g(x) &= (4x)^3 - 3 \times (4x)^2 \times 3 + 3 \times (4x) \times 3^2 - 3^3 \\ &= 64x^3 - 144x^2 + 108x - 27 \end{aligned}$$

$$f(x) + g(x)$$

$$= (4x^2 - 20x + 25) + (64x^3 - 144x^2 + 108x - 27)$$

$$= 64x^3 - 140x^2 + 88x - 2$$